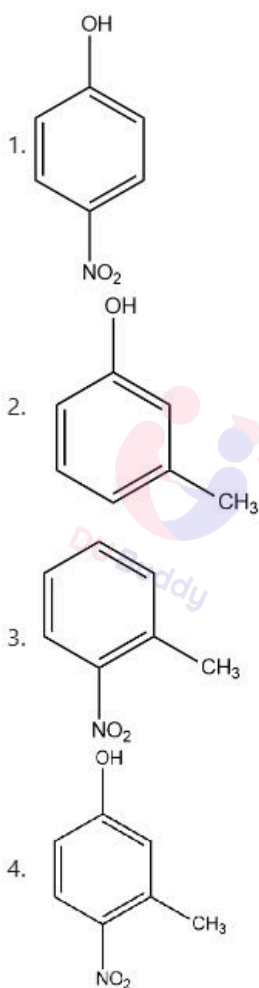


Q.1

Write the major product when 3-methylphenol is subjected to mononitration.



Options

1. 1
2. 2
3. 3
4. 4

Q.2 The number of peptide linkages in a tripeptide containing three amino acids is :

1. 3
2. 2
3. 4
4. 5

Options

1. 1
2. 2
3. 3
4. 4

Q.3 Cellulose is a _____ polymer composed of _____.

1. Linear, α -D-glucose
2. Branched, β -D-glucose
3. Linear, β -D-glucose
4. Branched, α -D-glucose

Options

1. 1
2. 2
3. 3
4. 4

Q.4 A first order reaction is found to have a rate constant, $k = 5.5 \times 10^{-14} \text{ s}^{-1}$. The half life of the reaction will be :

1. $1.26 \times 10^{11} \text{ s}$
2. $1.26 \times 10^{13} \text{ s}$
3. $1.26 \times 10^{19} \text{ s}$
4. $1.86 \times 10^{11} \text{ s}$

Options 1. 1

2. 2
3. 3
4. 4

Q.5 On what basis Scandium is considered as transition metal, but Zinc is not considered as transition metal?

1. presence of filled s orbital in Scandium and Zinc.
2. on the basis of an incompletely filled 3d orbital in the case of Scandium atom in its ground state, but Zinc has completely filled 3d orbital in its ground and its oxidation state.
3. Zinc shows d-d transition.
4. 3d orbital filled prior to 4s in Zinc

Options 1. 1

2. 2
3. 3
4. 4

Q.6 The steam distillation method is used for the separation of 1 : 1 mixture of p-nitrophenol and o-nitrophenols because

1. p-nitrophenol forms an intramolecular H-bond with the nitro group of the same molecule and o-nitrophenol forms an inter-molecular H-bond with the nitro group of other molecules.
2. p-nitrophenol and o-nitrophenol form an intramolecular H-bond with the nitro group of the same molecule.
3. p-nitrophenol forms an inter-molecular H-bond with the nitro group of the other molecules and o-nitrophenol forms an intramolecular H-bond with the nitro group of the same molecule.
4. p-nitrophenol and o-nitrophenol form an inter-molecular H-bond with the nitro group of the other molecules.

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Arrange the following in the increasing order of their spin-only magnetic moment values in free state

- (A) Mn^{3+}
(B) Fe^{3+}
(C) Sc^{3+}
(D) Cr^{3+}

Choose the correct answer from the options given below:

1. (C) (D) (B) (A)
2. (A) (C) (D) (B)
3. (C) (D) (A) (B)
4. (B) (A) (C) (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Isotonic solutions must have the same

- (A) Solute
(B) Density
(C) Elevation in boiling point
(D) Depression in freezing point

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (C) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Based on solute-solvent interaction, the solute which will be most soluble in n-octane is:

1. C_6H_{14}
2. KCl
3. CH_3CN
4. CH_3OH

Options 1. 1

2. 2

3. 3

4. 4

Q.10 Arrange the following in decreasing order of their basic strength in aqueous solution

- (A) $C_6H_5NH_2$
(B) $C_2H_5NH_2$
(C) $(C_2H_5)_2NH$
(D) NH_3

1. (C), (B), (D), (A)
2. (C), (B), (A), (D)
3. (B), (C), (D), (A)
4. (C), (D), (B), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Arrange the following mixtures in increasing order of their resultant molarity

- (A) 20 mL of 0.5 M HCl + 30 mL of 0.3 M HCl
(B) 30 mL of 0.3 M HCl + 30 mL of 0.3 M HCl
(C) 20 mL of 0.5 M HCl + 20 mL of 0.3 M HCl
(D) 20 mL of 0.3 M HCl + 30 mL of 0.5 M HCl

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (A), (B)
3. (B), (A), (C), (D)
4. (C), (A), (B), (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.12 The C-Br bond in bromobenzene as compared with C-Br bond in ethyl bromide is

1. Longer and weaker
2. Shorter and weaker
3. Shorter and stronger
4. Longer and stronger

Options 1. 1

2. 2

3. 3

4. 4

Q.13 Boiling of Sucrose with water in the presence of acid H^+ will give:

1. Glucose and Fructose
2. Cellulose and Glucose
3. Cellulose and Fructose
4. Starch and Glucose

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Match the compounds given in List-I with the oxidation state of cobalt present in it given in

List-II

List-I

Compound

List-II

Oxidation state of cobalt

List-I	List-II
(A) $[\text{Co}(\text{NCS})(\text{NH}_3)_5](\text{SO}_3)$	(I) +3
(B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{SO}_4$	(II) +2
(C) $\text{Na}_4[\text{Co}(\text{S}_2\text{O}_3)_3]$	(III) 0
(D) $\text{Co}_2(\text{CO})_8$	(IV) +4

Choose the correct answer from the options given below:

- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (I), (B) - (IV), (C) - (II), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Propene on addition with bromine in CCl_4 gives

- 1,2-Dibromopropane
- 2,2-Dibromopropane
- Propylbromide
- 2-Bromopropane

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Match List-I (vitamin) with List-II (deficiency disease)

List-I	List-II
Vitamin	Deficiency disease
(A) Thiamine	(I) Convulsions
(B) Pyridoxine	(II) Cheilosis
(C) Vitamin E	(III) Retarded Growth
(D) Riboflavin	(IV) Muscular weakness

Choose the correct answer from the options below:

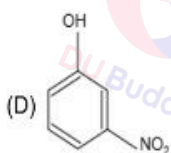
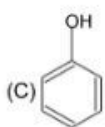
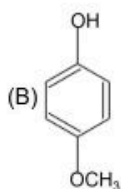
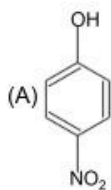
- (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
- (A) - (I), (B) - (III), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.17

The correct order of decreasing pK_a values of the following compounds



Choose the **correct** answer from the options given below:

1. (B) , (C), (D), (A)
2. (D), (A), (C), (B)
3. (C), (D), (A), (B)
4. (A) , (D), (C), (B)

Options 1. 1

2. 2

3. 3

4. 4

Q.18 The molality of a solution made by dissolving 2.5 g of ethanoic acid (CH_3COOH) in 75 g of benzene is:

1. $0.432 \text{ mol kg}^{-1}$
2. $0.556 \text{ mol kg}^{-1}$
3. $0.332 \text{ mol kg}^{-1}$
4. $0.752 \text{ mol kg}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Gabriel phthalimide synthesis method is used for the preparation of:

1. Secondary Amine
2. Primary Amine
3. Tertiary Amine
4. Quaternary amine

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Select the correct statements for Hoffmann bromamide degradation reaction:

- (A) Used to prepare primary amines

- (B) Amides are treated with bromine in aqueous solution.
 (C) NaOH is used in this reaction.
 (D) HCl is used in this reaction

Choose the correct answer from the options given below:

- (A), (B) and (D) only
- (A), (B) and (C) only
- (A), (B), (C) and (D)
- (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Identify the order of the two reactions whose rate constants are given below:

(i) $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$

(ii) $k = 3 \times 10^{-4} \text{ s}^{-1}$

- (i) Second order, (ii) Zero order
- (i) Second order, (ii) First order
- (i) Zero order, (ii) First order
- (i) First order, (ii) Second order

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Match the type of hybridisation given in List-I with their geometry given in List-II

List-I	List-II
Type of hybridisation	Geometry
(A) sp^3	(I) Trigonal bipyramidal
(B) dsp^2	(II) Octahedral
(C) sp^3d	(III) Square planar
(D) sp^3d^2	(IV) Tetrahedral

Choose the correct answer from the options below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
- (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Match List-I with List-II

List-I	List-II
Order of reaction	Expressions/Unit
(A) Unit of zero order	(I) $kt = [R]_0 - [R]_t$
(B) Unit of First order	(II) $\text{mol L}^{-1} \text{ s}^{-1}$
(C) Integrated Rate Law for zero order	(III) s^{-1}
(D) Integrated Rate Law for First order	(IV) $[R] = [R]_0 e^{-kt}$

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Number of unpaired electrons in the square planar $[\text{Pt}(\text{CN})_4]^{2-}$ are:

1. Two unpaired electrons
2. Zero unpaired electron
3. One unpaired electrons
4. Three unpaired electrons

Options 1. 1

2. 2
3. 3
4. 4

Q.25 The correct order of increasing reactivity of C-X bond towards nucleophilic substitution reaction ($\text{S}_{\text{N}}1$) in the following compounds is

- (A) Chlorobenzene
- (B) Benzyl Chloride
- (C) Tertiary Butyl Chloride
- (D) 1-Chlorobutane

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (D), (C), (B)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Name the reaction when a mixture of acetaldehyde and propanal is treated with aqueous alkali solution.

1. Cross Aldol Condensation
2. Self Aldol Condensation
3. Intramolecular Aldol Condensation
4. Cannizzaro Reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Which of the following is not a limitation of valence bond theory (VBT)?

1. It does not distinguish between weak and strong ligands.
2. It does not explain the color exhibited by the coordination complex.
3. It does not explain the concept of the high spin and low spin complex.
4. It does not give predictions regarding the tetrahedral and square plane complex in coordination number 4.

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Isopropylbromide reacts with silver nitrite to form major product as

1. 2-Nitropropane
2. Propane-2-nitrite
3. Propane -2-nitrile
4. Propane

Options 1. 1

2. 2
3. 3
4. 4

Q.29 The IUPAC name of $[\text{CoCl}_2(\text{en})_2]\text{Cl}$ complex is:

1. Dichloridobis(ethane-1,2-diamine)cobalt(III) chloride
2. Dichloridobis(ethane-1,2-diamine)cobalt(II) chloride
3. Bis(ethane-1,2-diamine)dichloridocobalt(II) chloride
4. Bis(ethane-1,2-diamine)dichloridocobalt(III) chloride

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Reaction of Chloroethane with ethanolic ammonia will result in:

1. Ethanamine
2. N-Ethylethanamine
3. N,N-Diethylethanamine
4. Quaternary ammonium salt

Options

1. 1
2. 2
3. 3
4. 4

Q.31 Choose the incorrect statements from the following:

- (A) The dipole moment of cyclohexyl chloride is smaller than chlorobenzene.
- (B) Alkyl halides are immiscible with water.
- (C) The dipole moment of methyl chloride is higher than that of methyl fluoride.
- (D) Chloroform doesn't undergo aerial oxidation in the presence of light

Choose the correct answer from the options given below:

1. (A), (B) and (D) only.
2. (B), (C) and (D) only.
3. (A), (B), (C) and (D).
4. (A), (C) and (D) only.

Options

1. 1
2. 2
3. 3
4. 4

Q.32 Which of the following statements is correct for an enzyme?

- (A) It is a biocatalyst.
- (B) It can catalyze any chemical reaction.
- (C) Almost all enzymes are globular proteins.
- (D) They reduce the magnitude of activation energy.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only.
2. (A), (C) and (D) only.
3. (A), (B), (C) and (D).
4. (B), (C) and (D) only.

Options

1. 1
2. 2
3. 3
4. 4

Q.33 45 g of ethylene glycol ($C_2H_6O_2$) is mixed with 600 g of water. Calculate the depression in freezing point.

(K_f for water = $1.86 \text{ K kg mol}^{-1}$)

1. 2.25 K
2. 2.32 K
3. 2.7 K
4. 2.9 K

Options

1. 1
2. 2
3. 3
4. 4

Q.34 According to the collision theory which one of the following statements is correct?

1. The colliding molecules should have proper orientation and at least threshold energy for the reaction to occur.
2. The colliding molecules should have proper orientation and energy less than a threshold energy for the reaction to occur.
3. If the colliding molecules have proper orientation then the reaction will always occur.
4. The rate of the reaction is exactly equal to the number of collisions taking place per unit time

Options

1. 1
2. 2
3. 3
4. 4

Q.35 The correct 4f sub-shell electronic configuration of

Yb^{2+} ion is

1. $4f^{14}$
2. $4f^{13}$

3. $4f^{12}$

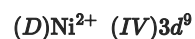
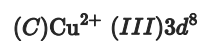
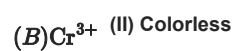
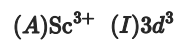
4. $4f^7$

Options

1. 1
2. 2
3. 3
4. 4

Q.36 Match the metal/ion given in List-I with properties given in List-II

List-I	List-II
Metal	Properties



Choose the correct answer from the options given below:

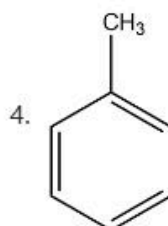
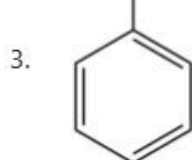
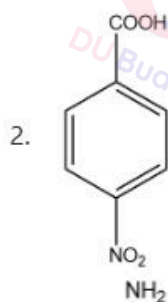
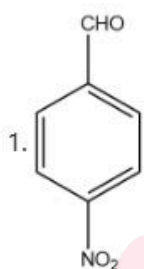
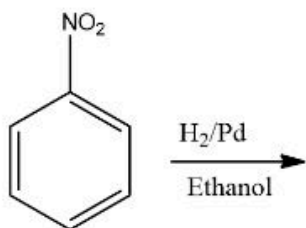
1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.37

Identify the product of the following reaction.



Options

1. 1
2. 2
3. 3
4. 4

Q.38 Out of the following given statements, identify the correct ones:

- (A) Yttrium does not exhibit variable oxidation states, yet it is a transition element.
- (B) Many compounds of transition elements act as catalysts in various chemical reactions.
- (C) Technetium is an artificially synthesized element of 4d transition series.
- (D) The E° value for the Mn^{3+}/Mn^{2+} couple is much more positive than that for Cr^{3+}/Cr^{2+} .

Choose the correct answer from the options given below:

1. (A), (B) and (C) only.
2. (A), (B) and (D) only.
3. (A), (B), (C) and (D).
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.39 The cleavage of methyl tertbutyl ether with HI gives

1. methanol and tertbutyl iodide
2. methyl iodide and tertbutanol
3. tertbutyl iodide
4. methanol

Options 1. 1

2. 2

3. 3

4. 4

Q.40

Identify the correct expression of Arrhenius equation:

1. $k = Ae^{-E_a/RT}$
2. $k = Ae^{E_a/RT}$
3. $k = Ae^{E_a/T}$
4. $k = Ae^{E_a/R}$

Options 1. 1

2. 2

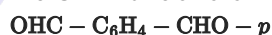
3. 3

4. 4

Q.41 Read the passage carefully and answer the Questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

The IUPAC name of following compound is



1. Benzene-1,4-dial
2. 1,4-Dioxobenzene
3. 4-Formylbenzaldehyde
4. Benzene-1,4-dicarbaldehyde

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the passage carefully and answer the Questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

Arrange the following in the increasing order of their pK_a values:

- (A) $\text{CF}_3\text{-COOH}$
- (B) $\text{O}_2\text{N-CH}_2\text{-COOH}$
- (C) $\text{NC-CH}_2\text{-COOH}$

(D) H-COOH

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (B), (A)
3. (A), (C), (B), (D)
4. (D), (B), (C), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the passage carefully and answer the Questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

Propyne on hydration in presence of mercury ions produce

1. Acetone
2. Propanal
3. 1-Propanol
4. Propan-2-ol

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the passage carefully and answer the Questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

The products formed when 2,2-Dimethylpropan-1-ol is heated with concentrated KOH are:

1. Potassium 2,2-Dimethylpropan-1-oate and 2,2-Dimethylpropan-1-ol
2. 2,2-Dimethylprop-1-ene and 2,2-Dimethylpropan-1-oate
3. 2,2-Dimethylpropan-1-ol and Potassium 2-Methylbutan-1-oate
4. Potassium 2,2-Dimethylpropan-1-oate and 3-Methylbutan-2-one

Options 1. 1

2. 2

3. 3

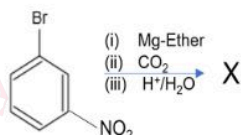
4. 4

Q.45 Read the passage carefully and answer the Questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the

carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

Identify the final product (X) of the chemical reaction obtained from the following transformation:



1. 3-Nitrobenzaldehyde
2. 1-Methyl-3-nitrobenzene
3. 3-Nitrobenzoic acid
4. 3-Aminobenzoic acid

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the passage carefully and answer the Questions

Molar conductivity (Λ_m) of a solution at a given concentration (c) is the conductance of volume, V of solution, containing one mole of electrolyte kept between the two electrodes with area of cross-section A and at a distance of unit length. It increases with the decrease in concentration and when the concentration approaches zero, the molar conductivity is called limiting molar conductivity (Λ_m^0). For a strong electrolyte, Λ_m increases linearly with dilution and is given by $\Lambda_m = \Lambda_m^0 - Ac^{1/2}$. The value of the constant A for a given solvent depends on the type of electrolyte along with temperature. According to Kohlrausch law, the value of Λ_m^0 for an electrolyte is $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$, where ν_+ and ν_- are the number of cations and anions, respectively, per molecule of the electrolyte and λ_+^0 and λ_-^0 are limiting molar conductivities of cation and anion, respectively. Kohlrausch law finds many applications, like determining the solubility of a sparingly soluble salt, determining the degree of dissociation (Λ_m/Λ_m^0), and the dissociation constant of a weak electrolyte.

If the molar conductance values of Ca^{2+} and Cl^- at infinite dilution are, respectively, $118.8 \times 10^{-4} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$ and $77.33 \times 10^{-4} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$, then the molar conductance of CaCl_2 at infinite dilution will be:

1. $273.46 \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$
2. $196.13 \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$
3. $41.47 \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$
4. $-41.47 \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Read the passage carefully and answer the Questions

Molar conductivity (Λ_m) of a solution at a given concentration (c) is the conductance of volume, V of solution, containing one mole of electrolyte kept between the two electrodes with area of cross-section A and at a distance of unit length. It increases with the decrease in concentration and when the concentration approaches zero, the molar conductivity is called limiting molar conductivity (Λ_m^0). For a strong electrolyte, Λ_m increases linearly with dilution and is given by $\Lambda_m = \Lambda_m^0 - Ac^{1/2}$. The value of the constant A for a given solvent depends on the type of electrolyte along with temperature. According to Kohlrausch law, the value of Λ_m^0 for an electrolyte is $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$, where ν_+ and ν_- are the number of cations and anions, respectively, per molecule of the electrolyte and λ_+^0 and λ_-^0 are limiting molar conductivities of cation and anion, respectively. Kohlrausch law finds many applications, like determining the solubility of a sparingly soluble salt, determining the degree of dissociation (Λ_m/Λ_m^0), and the dissociation constant of a weak electrolyte.

The plot between Λ_m and $c^{1/2}$ is a straight line with

1. Slope = Λ_m^0 and intercept = A
2. Slope = Λ_m^0 and intercept = $-A$
3. Slope = A and intercept = Λ_m^0
4. Slope = $-A$ and intercept = Λ_m^0

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the passage carefully and answer the Questions

Molar conductivity (Λ_m) of a solution at a given concentration (c) is the conductance of volume, V of solution, containing one mole of electrolyte kept between the two electrodes with area of cross-section A and at a distance of unit length. It increases with the decrease in concentration and when the concentration approaches zero, the molar conductivity is called limiting molar conductivity (Λ_m^0). For a strong electrolyte, Λ_m increases linearly with dilution and is given by $\Lambda_m = \Lambda_m^0 - Ac^{1/2}$. The value of the constant A for a given solvent depends on the type of electrolyte along with temperature. According to Kohlrausch law, the value of Λ_m^0 for an electrolyte is $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$, where ν_+ and ν_- are the number of cations and anions, respectively, per molecule of the electrolyte and λ_+^0 and λ_-^0 are limiting molar conductivities of cation and anion, respectively. Kohlrausch law finds many applications, like determining the solubility of a sparingly soluble salt, determining the degree of dissociation (Λ_m/Λ_m^0), and the dissociation constant of a weak electrolyte.

Which one of the following has a different value of constant A in water?

1. CaCl_2

2. KCl

3. NaCl

4. LiCl

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the passage carefully and answer the Questions

Molar conductivity (Λ_m) of a solution at a given concentration (c) is the conductance of volume, V of solution, containing one mole of electrolyte kept between the two electrodes with area of cross-section A and at a distance of unit length. It increases with the decrease in concentration and when the concentration approaches zero, the molar conductivity is called limiting molar conductivity (Λ_m^0). For a strong electrolyte, Λ_m increases linearly with dilution and is given by $\Lambda_m = \Lambda_m^0 - Ac^{1/2}$. The value of the constant A for a given solvent depends on the type of electrolyte along with temperature. According to Kohlrausch law, the value of Λ_m^0 for an electrolyte is $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$, where ν_+ and ν_- are the number of cations and anions, respectively, per molecule of the electrolyte and λ_+^0 and λ_-^0 are limiting molar conductivities of cation and anion, respectively. Kohlrausch law finds many applications, like determining the solubility of a sparingly soluble salt, determining the degree of dissociation (Λ_m/Λ_m^0), and the dissociation constant of a weak electrolyte.

The unit of constant A is:

1. $\text{S cm}^2 \text{ mol}^{-1}$.

2. $\text{S cm}^2 \text{ mol}^{-1} \text{ L}^{-1}$.

3. $\text{S cm}^2 \text{ mol}^{-3/2} \text{ L}^{1/2}$.

4. $\text{S cm}^2 \text{ mol}^{-1/2} \text{ L}^{-1/2}$.

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the passage carefully and answer the Questions

Molar conductivity (Λ_m) of a solution at a given concentration (c) is the conductance of volume, V of solution, containing one mole of electrolyte kept between the two electrodes with area of cross-section A and at a distance of unit length. It increases with the decrease in concentration and when the concentration approaches zero, the molar conductivity is called limiting molar conductivity (Λ_m^0). For a strong electrolyte, Λ_m increases linearly with dilution and is given by $\Lambda_m = \Lambda_m^0 - Ac^{1/2}$. The value of the constant A for a given solvent depends on the type of electrolyte along with temperature. According to Kohlrausch law, the value of Λ_m^0 for an electrolyte is $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$, where ν_+ and ν_- are the number of cations and anions, respectively, per molecule of the electrolyte and λ_+^0 and λ_-^0 are limiting molar conductivities of cation and anion, respectively. Kohlrausch law finds many applications, like determining the solubility of a sparingly soluble salt, determining the degree of dissociation (Λ_m/Λ_m^0), and the dissociation constant of a weak electrolyte.

If the equivalent conductivity of 1 M Phthalic acid is $12.8 \text{ S cm}^2 \text{ eq}^{-1}$ and the limiting equivalent conductivities of the

hydrogen and Phthalate ions are 42 and $288.42 \text{ S cm}^2 \text{ eq}^{-1}$, respectively, the degree of dissociation of Phthalic acid will be:

1. 0.304
2. 0.044
3. 0.387
4. 0.0387

Options

1. 1
2. 2
3. 3
4. 4